Predicting Lung Cancer Survival Outcomes Using Machine Learning

Combined Report : Unit 1 - Unit 4

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Date: October 2026

1. Introduction

Lung cancer is one of the most serious types of cancer and a major global health problem. Even with significant advances in treatment like immunotherapy and other advanced therapies, survival rates still vary widely among patients. Right now, doctors estimate prognosis by looking at how advanced the cancer is and drawing on their clinical experience. But these methods may not give a complete picture of an individual patient's situation, which can lead to treatment decisions that aren't optimal for that person.

What makes lung cancer particularly challenging is that survival varies widely even among patients with the same stage, which makes it a good candidate for machine learning analysis. ML offers a way to analyze large amounts of patient data and find patterns that humans might miss. This project explores whether ML models can predict lung cancer survival using patient demographics and clinical factors.

I'm interested in applying data science to healthcare. I want to understand how these tools work and whether they could realistically help doctors make better decisions. This project is an opportunity to apply what I've learned in my coursework to a meaningful problem.

2. Business Problem and Research Objectives

The Business Problem

Predicting survival outcomes for lung cancer patients is challenging for healthcare providers. Doctors often rely on cancer stage, patient characteristics, treatment information, and clinical experience, but these factors don't always provide an accurate picture of a patient's expected outcome. From a healthcare analytics perspective, better survival predictions could help hospitals

and healthcare organizations with treatment planning, resource allocation, and identifying patients who may need additional support. It could also give patients and their families more useful information about their diagnosis. The goal is not to replace doctors' decisions, but to explore whether data-driven models can provide additional information to support healthcare decision-making.

Research Objectives

Primary objective: Build and evaluate machine learning models that can predict survival outcomes for lung cancer patients using demographic and clinical data.

Secondary objectives:

1. Compare the performance of different machine learning algorithms (Random Forest, XGBoost, Logistic Regression, SVM).
2. Identify which patient characteristics are the strongest predictors of survival.
3. Assess whether these models could be useful in clinical settings.
4. Compare ML models against simpler baseline approaches.

Expected Contributions

Methodologically, I will document a reproducible approach for building survival prediction models using publicly available data—a framework that could be adapted for other similar research. Clinically, if the models perform well, they could eventually help doctors make more informed decisions; at minimum, identifying key prognostic factors could guide clinical attention. Practically, this project will help me develop skills in healthcare data science, including preprocessing clinical data, building ML models, and interpreting results—experience that will be valuable for my career goals.

3. Dataset Description

Where the Data Came From

I used the Kaggle "Lung Cancer Prediction" dataset created by Rashad Mammadov. It's publicly available and contains demographic, clinical, treatment, and survival information for lung cancer

patients. A 2025 published study also used this dataset for lung cancer survival prediction and reported 23,658 patient records (Mammadov, 2025).

What's in the Data

- Sample size: 23,658 patient records (48 columns before engineered features were added; 57 columns in the working dataset)
- Features: Demographic and clinical information such as age, gender, race/ethnicity, cancer stage, tumor size, blood pressure, laboratory values (hemoglobin, white blood cell count, platelet count, albumin, alkaline phosphatase, liver enzymes, creatinine, LDH, calcium, phosphorus, glucose, potassium, sodium), smoking history, treatment type, comorbidities, insurance type, and family history
- Class balance: Nearly balanced classification problem

The Target Variable

The prediction target is a binary survival outcome derived from Survival_Months using a 60-month cutoff:

Survival Group	Definition	Patients	Percentage
Long-Term Survivor	Survival_Months ≥ 60	11,948	50.50%
Short-Term Survivor	Survival_Months < 60	11,710	49.50%

The dataset is well-balanced across classes and suitable for binary classification.

Why I Chose This Dataset

I chose it because it's publicly available and large enough for a student machine learning project. It also contains demographic and clinical information that matches the research questions in this project, and it has been used in previous research, which provides a useful reference when comparing my results.

Alternative Data Sources

If the primary dataset turned out not to be suitable after further analysis, possible alternatives included:

- **SEER Database:** A large cancer registry with extensive information on cancer cases and survival outcomes.
- **TCGA-LUAD:** A lung adenocarcinoma dataset that includes genomic and clinical information.
- **UCI Thoracic Surgery Dataset:** A smaller dataset that could be used as an alternative for comparison.

4. Preliminary Literature Review

Previous research shows that machine learning can be useful for predicting lung cancer survival. Altuhaifa et al. (2023) reviewed different machine learning approaches and found that factors such as age, gender, and cancer stage were commonly important predictors. Didier et al. (2024) also examined machine learning approaches for lung cancer survival prediction and found that ML can provide strong predictive performance, although more validation is needed before these models can be widely used in clinical practice.

Other research has compared machine learning methods with traditional survival models such as Cox regression. Germer et al. (2024) compared Cox regression with several machine learning models and found that performance varied depending on the clinical information available. Similarly, Huang et al. (2025) reviewed studies comparing machine learning with Cox regression and found that machine learning did not consistently perform better than traditional survival methods. Recent research has also examined models such as Random Forest, XGBoost, and Support Vector Machines for lung cancer prediction. These studies suggest that machine learning can identify useful patterns in clinical data, but model interpretation remains an important challenge.

Interpretability is especially important in healthcare because users need to understand why a model makes a particular prediction. Lundberg and Lee (2017) introduced SHAP, which can be used to explain the contribution of individual features to machine learning predictions. Overall, previous research supports the potential of machine learning for lung cancer survival prediction, but results vary across datasets and models. This project builds on that research by comparing several machine

learning models using demographic and clinical data and identifying the factors that are most important for predicting survival outcomes.

5. Data Quality Assessment

I ran a thorough quality assessment on all 38 original variables, and the results were a nice surprise:

Check	Result
Missing values	0 across all 38 variables
Duplicate rows	0
Duplicate Patient_IDs	0
Data type issues	None
Inconsistent categorical values	None detected

Summary statistics (df.describe().T) confirmed that all numerical features fall within clinically plausible ranges, there were no constant or near-constant columns, and no variables exhibited extreme skew requiring transformation. The average patient age was about 54.4 years (SD 14.4), average tumor size was about 55.4 mm (SD 26.0), and mean systolic and diastolic blood pressure were approximately 134.5 and 84.5, respectively.

Missing values: Every variable reported Missing_Count = 0 and Missing_Percentage = 0.0%. No missing-value treatment was performed—fabricating imputation when no data is missing would be methodologically incorrect.

Outliers: I applied the IQR method (1.5 × IQR rule) to all numerical variables and identified zero outliers. No records were removed. Outlier detection was documented rather than "fixed"—the correct procedure when no outliers exist.

6. Data Cleaning and Preprocessing

Data Cleaning Procedures

1. **Removed** Patient_ID—an identifier with no predictive value.
2. **Removed** Survival_Months **from predictors**—it was used to construct the target, so retaining it would cause target leakage.
3. **Dropped redundant raw text columns** (e.g., "Gender," "Stage," "Treatment," "Smoking_History"). The dataset already contained one-hot encoded versions of these columns, so keeping both would have duplicated the same information and confused the models.
4. Confirmed 0 missing values and 0 duplicates post-cleaning.

Feature Encoding

Categorical predictor variables were one-hot encoded using pd.get_dummies(drop_first=True, dtype=int). The encoding was fitted on the training set only, then applied to the test set, with test columns reindexed to match training columns exactly. Using drop_first=True avoids multicollinearity (the dummy variable trap). I chose one-hot encoding over label encoding because all categorical variables are nominal—there's no inherent ordering.

Feature Scaling

Numerical features were standardized using StandardScaler (mean = 0, std = 1). The scaler was fitted on the training set only, and the same transformation was applied to the test set. This prevents information leakage from the test set into the training pipeline. I chose StandardScaler because the downstream models are distance-based (logistic regression, KNN, SVM).

Train/Test Split

The split was performed before encoding and scaling, using an **80/20 stratified split** (random_state=42) to preserve class proportions. This resulted in 18,926 training rows and 4,732

test rows, with the target proportions preserved at 50.5% Long-Term and 49.5% Short-Term in both sets.

Final Cleaned Dataset

- **Rows:** 23,658
- **Columns: 57 in the working dataset**
- **Predictors used for modeling: 52 total — 50 numeric predictors plus 2 categorical predictors (Age_Group, Tumor_Size_Group)**
- **Predictors used for Variance Threshold / PCA: 50 numeric predictors only (the two categorical predictors were retained as-is and not fed into PCA)**
- **Missing values:** 0
- **Duplicates:** 0
- **Target:** Survival_Group (Short-Term / Long-Term)
- **File:** data/cleaned_dataset_assignment3.csv

Reflection on Preprocessing Decisions

The most consequential decisions were (1) removing Survival_Months to prevent target leakage, and (2) performing the train/test split before encoding and scaling. Both are standard best practices but easy to overlook. Given the absence of missing values and outliers, the correct response was to document those findings rather than force unnecessary preprocessing steps. The resulting cleaned dataset is directly usable in the modeling stage without further transformation.

7. Exploratory Data Analysis

Target Distribution

The target variable Survival_Group was almost perfectly balanced—50.5% Long-Term and 49.5% Short-Term. This was great because it meant I didn't have to use techniques like SMOTE to fix class imbalance, and I could rely on accuracy and macro-F1 as my main metrics.

Correlation Analysis

Pearson correlation analysis found no predictor pairs above the notebook's selected absolute correlation threshold of 0.70. The strongest listed correlations were mostly between one-hot encoded categories (for example, different tumor-location or smoking-history categories), which are expected to be negatively related when categories are mutually exclusive. Individual numeric predictors had very small correlations with the survival target; the largest displayed target correlation was about 0.013 in absolute value. Correlation alone therefore did not identify a strong single predictor of survival—a red flag for the modeling phase.

Five Business Insights Supported by Visual Analysis

1. **Balanced outcome:** Long-Term survival represents 50.50% and Short-Term survival 49.50% of the dataset, so there is no major class imbalance.
2. **Age:** The mean age is approximately 54.5 years for Long-Term patients and 54.4 years for Short-Term patients. The overlapping distributions suggest age alone may not separate the groups well.
3. **Tumor size:** Mean tumor size is approximately 55.6 mm for Long-Term patients and 55.1 mm for Short-Term patients. The small difference and overlapping distributions suggest limited separation by tumor size alone.
4. **Cancer stage:** The stage count plot shows broadly similar outcome distributions across stages in this dataset; stage alone does not create a strong visual separation in the observed data.
5. **Weak individual correlations:** Numeric predictor correlations with the encoded survival outcome are small, suggesting that a useful model may need to combine multiple variables and capture patterns not represented by simple linear correlations.

Engineered Features

I engineered five new features to provide additional ways to examine the data:

Feature	How It Was Created	Why It Was Added
Age_Group	Age grouped into 30–49, 50–64, and 65–79 bands	To compare survival patterns across broader age ranges

Tumor_Size_Group	Tumor size grouped as Small (up to 40 mm), Medium (40–70 mm), and Large (over 70 mm)	To make tumor-size patterns easier to compare categorically
Comorbidity_Count	Sum of seven yes/no comorbidity indicators	To summarize the number of recorded comorbidities for each patient
Treatment_Count	Sum of three treatment indicator columns	To create a compact count from the treatment indicators; interpretation depends on how those indicators are encoded
Blood_Pressure_Difference	Systolic blood pressure minus diastolic blood pressure	To represent the difference between the two blood-pressure measurements

I added these partly just to see if they would help the models.

Variance Threshold

I applied a Variance Threshold with a threshold of 0.01 to the 50 numeric predictors (the two categorical predictors, Age_Group and Tumor_Size_Group, were not part of this step). The recorded output retained all 50 numeric predictors, so this filter did not reduce the feature count. This suggests that none of the included predictors fell below the selected variance threshold.

Principal Component Analysis (PCA)

PCA was run on the 50 numeric predictors only, since PCA requires numeric inputs and the two categorical predictors were already represented as compact grouped variables. In the saved notebook outputs, PCA required 39 components to retain 90% of the variance and 42 components to retain 95% of the variance, compared with 50 original numeric predictors. At the 90% level, that's 11 fewer components—a 22% reduction in feature count.

Original features versus PCA features (Logistic Regression):

Approach	Features	Logistic Regression Accuracy
Original numeric Features	50	49.79%
PCA Features (90% variance)	39	49.64%

The recorded accuracies are very similar. These results are close to chance-level accuracy for a nearly balanced two-class target, so this simple Logistic Regression model should not be presented as a strong predictor. PCA reduced dimensionality with only a small difference in accuracy, but principal components are harder to interpret than the original variables.

EDA Reflection

EDA helped me understand the structure of the dataset before modeling. The plots and correlation analysis showed that the survival groups were balanced and that individual variables did not strongly separate the outcomes. This encouraged me to engineer grouped and summary features and to test a dimensionality-reduction method instead of assuming that one variable would explain survival.

8. Baseline Algorithms

I compared three classifiers. I used Logistic Regression as a linear baseline with a maximum of 2,000 iterations. I set the Decision Tree Classifier to a maximum depth of 8 and a minimum leaf size of 20 to limit tree complexity. For K-Nearest Neighbors, I used 15 neighbors with distance weighting and applied scaling because KNN depends on distances between observations. These settings are baseline configurations, not a comprehensive hyperparameter search.

9. Evaluation Metrics

I evaluated all models on the same held-out test set. Accuracy measures the proportion of correct predictions. Macro precision, recall, and F1 average the corresponding scores across the two classes with equal class weight. Class-specific F1 scores show the balance between precision and recall for Short-Term and Long-Term separately. I calculated ROC-AUC using Long-Term as the positive class; an AUC near 0.50 indicates little ability to rank the two classes above chance. The confusion matrices and class-specific classification reports are in the accompanying notebook.

10. Results and Model Comparison

Model	Accuracy	Macro precision	Macro recall	Macro F1	Long-Term F1	Short-Term F1	ROC-AUC
Logistic Regression	0.498	0.498	0.498	0.496	0.537	0.454	0.503
Decision Tree	0.494	0.491	0.492	0.476	0.573	0.379	0.498
K-Nearest Neighbors	0.493	0.493	0.493	0.493	0.500	0.487	0.498

Logistic Regression achieved the highest test accuracy (0.499) and macro-F1 (0.496) among the three baseline models. KNN's macro-F1 was 0.493, while the Decision Tree's macro-F1 was 0.476. However, all test accuracies were approximately 0.49–0.50 and all ROC-AUC values were close to 0.50. These values indicate performance close to chance rather than convincing discrimination between the survival groups. I don't interpret the small numerical differences as evidence of a reliable predictive advantage.

I selected Logistic Regression as the leading baseline by macro-F1, not as a clinically suitable model. Its Long-Term recall was 0.575, compared with Short-Term recall of 0.421, showing that the model detected the Long-Term class more often but missed a substantial proportion of Short-Term cases. The Decision Tree showed an even larger class-specific recall difference. KNN had more similar class recall but still produced weak overall discrimination.

11. Cross-Validation and Generalization

I ran three-fold training cross-validation for each model:

- Logistic Regression: macro-F1 scores were 0.492, 0.497, 0.508 (mean = 0.499, SD = 0.007).
- Decision Tree: macro-F1 scores were 0.485, 0.481, 0.469 (mean = 0.478, SD = 0.007).
- K-Nearest Neighbors: macro-F1 scores were 0.506, 0.500, 0.502 (mean = 0.503, SD = 0.002).

Cross-validation scores were also close to 0.50. This supports my conclusion that the current baseline feature set does not provide strong separation in this experiment. Cross-validation and a single split do not replace external validation, and I can't draw causal conclusions from these results.

12. Error Analysis: Five Misclassified Test Cases

The five cases below are actual misclassifications made by the model with the highest macro-F1 (Logistic Regression). I assigned case numbers for this report only; they are not patient identifiers. The feature summaries describe observations but do not establish why a prediction was wrong.

Case	Actual	Predicted	Age	Tumor size (mm)	Performance status	Comorbidity count

1	Long-Term	Short-Term	47	14.6	1	3
2	Long-Term	Short-Term	65	53.0	2	4
3	Long-Term	Short-Term	58	90.7	3	3
4	Short-Term	Long-Term	55	93.7	2	3
5	Short-Term	Long-Term	33	88.0	3	5

Case 1: actual class Long-Term, predicted class Short-Term. The recorded age was 47, tumor size was 14.6 mm, performance status was 1, stage indicators were summarized as Stage III, and comorbidity count was 3. These fields alone don't establish a clear reason for the error. The mistake may reflect overlapping feature patterns, missing predictive information, noise in the target label, or limitations of the model. I would need more complete data and independent validation to investigate the cause.

Case 2: actual class Long-Term, predicted class Short-Term. The recorded age was 65, tumor size was 53.0 mm, performance status was 2, stage indicators were summarized as Stage III, and comorbidity count was 4. These fields alone don't establish a clear reason for the error. The mistake may reflect overlapping feature patterns, missing predictive information, noise in the target label, or limitations of the model. I would need more complete data and independent validation to investigate the cause.

Case 3: actual class Long-Term, predicted class Short-Term. The recorded age was 58, tumor size was 90.7 mm, performance status was 3, stage indicators were summarized as Stage IV, and comorbidity count was 3. These fields alone don't establish a clear reason for the error. The mistake may reflect overlapping feature patterns, missing predictive information, noise in the target label, or limitations of the model. I would need more complete data and independent validation to investigate the cause.

Case 4: actual class Short-Term, predicted class Long-Term. The recorded age was 55, tumor size was 93.7 mm, performance status was 2, stage indicators were summarized as Stage IV, and comorbidity count was 3. These fields alone don't establish a clear reason for the error. The mistake may reflect overlapping feature patterns, missing predictive information, noise in the target label, or limitations of the model. I would need more complete data and independent validation to investigate the cause.

Case 5: actual class Short-Term, predicted class Long-Term. The recorded age was 33, tumor size was 88.0 mm, performance status was 3, stage indicators were summarized as Stage III, and comorbidity count was 5. These fields alone don't establish a clear reason for the error. The mistake may reflect overlapping feature patterns, missing predictive information, noise in the target label, or limitations of the model. I would need more complete data and independent validation to investigate the cause.

13. Limitations

First, all three models performed close to chance on the held-out test set, so the current feature set does not support strong individual-level prediction. Second, my analysis depends on how the binary survival groups were defined; I should document and verify the underlying time horizon and label construction. Third, some engineered variables duplicate or summarize existing measurements, and encoded and derived variables may introduce redundancy. Fourth, my evaluation uses one dataset and one held-out split; it does not demonstrate generalizability to other hospitals or patient populations. Fifth, feature importance and classification metrics are predictive summaries, not evidence that a feature causes survival outcomes. Finally, this project is educational and not a clinically validated decision-support system.

14. Future Recommendations

For future work, I would first verify the definition and reliability of Survival_Group and audit every feature for availability at prediction time. Next, I would establish a majority-class baseline and compare it with the current models. I would conduct hyperparameter tuning inside cross-validation on training data, with the test set reserved for final evaluation. I would review the feature engineering to reduce redundancy and preserve clinically meaningful information. I would expand error analysis across both classes and, where appropriate, across relevant subgroups while respecting privacy. If suitable data are available, I would evaluate the final model on an independent dataset and examine probability calibration before making claims about risk. Any clinical application would require expert review, formal validation, fairness assessment, and applicable regulatory and ethical safeguards.

15. Conclusion

I implemented three baseline classifiers with leakage-aware preprocessing and evaluated them using a stratified 80/20 split. Logistic Regression produced the highest macro-F1 among the evaluated baselines, but its performance and the ROC-AUC values were close to chance. My main finding is therefore not that I developed a successful survival predictor, but that these baseline models did not show reliable discrimination using the current dataset and settings. Data-definition checks, better feature validation, training-only tuning, and independent testing are necessary next steps.

References

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